

Revised Load Interconnection System Impact Study Report

For Queue Project Q23-2333

Ada Carbon Solutions Red River 138 kV
August 02, 2025

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q23-2333. The Interconnection Customer proposes a 111.1 MW large load (data center) facility interconnecting at the Ada Carbon Solutions Red River 138 kV substation in the Virginia area. The request was received on July 09, 2023. The study evaluated system intact and single contingency conditions using the regional planning case, including 381 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 8 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$18,066,000.

3. Network Impacts

3.1 Load Deliverability Analysis

1. Facility: CUMBERLAND-ARAPAHOE 138kV
Contingency: ROGERSVILLE-MT AIRY 138kV
Loading: 116.1% (Rating: 216 MVA)
DFAX: 60.0%
Pre-project loading: 85.3%
2. Facility: SHORT HILLS-ADA CARBON SOLUTIONS RED RIVER 138kV
Contingency: AGUAS BUENAS GIS TC-SANTA ANA 3 138kV
Loading: 109.0% (Rating: 166 MVA)
DFAX: 100.0%
Pre-project loading: 42.0%
3. Facility: PALERMO-SEPLUVEDA 138kV Cir. 1
Contingency: PALRMO-SEPLUVEDA 138kV Ckt 2
Loading: 106.6% (Rating: 286 MVA)
DFAX: 50.9%
Pre-project loading: 86.8%
4. Facility: SEPLUVEDA SUBSTATION-PALERMO 138kV Ckt 2
Contingency: SEPLUVEDA-PALERMO SUBSTATION 138kV Circuit 1
Loading: 104.5% (Rating: 291 MVA)
DFAX: 50.9%
Pre-project loading: 85.1%
5. Facility: ARAPAHOE-LOOSE CRK 138kV
Contingency: AGUAS BUENAS GIS TC-SANTA ANA 3 138kV Line
Loading: 103.7% (Rating: 206 MVA)
DFAX: 100.0%
Pre-project loading: 49.8%

6. Facility: SHORT HILLS-LOOSE CREEK 138kV
 Contingency: SANTA ANA 3-AGUAS BUENAS GIS TC 138kV L
 Loading: 103.7% (Rating: 206 MVA)
 DFAX: 100.0%
 Pre-project loading: 49.8%
7. Facility: NEPTUNE-FOREST GLEN 138kV
 Contingency: NEPTUNE 345 kV/138 kV XFMR
 Loading: 100.5% (Rating: 275 MVA)
 DFAX: 6.3%
 Pre-project loading: 98.0%
8. Facility: KAW-SPALL NS 138kV
 Contingency: WHATSHAN-FOX HILLS 138kV line
 Loading: 100.4% (Rating: 253 MVA)
 DFAX: 5.1%
 Pre-project loading: 98.1%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
CUMBERLAND-ARAPAHOE 138kV	Rebuild line with higher-capacity conductor	\$4,649,000	100% (\$4,649,000)
SHORT HILLS-ADA CARBON SOLUTIONS RED RIVER 138kV Line	Reconductor line	\$2,317,000	100% (\$2,317,000)
PALERMO-SEPLUVEDA 138kV Circuit 1	Reconductor line	\$3,546,000	100% (\$3,546,000)
SEPLUVEDA-PALERMO 138kV Ckt 2	Reconductor line	\$3,423,000	100% (\$3,423,000)
ARAPAHOE-LOOSE CRK 138kV	Reconductor line	\$3,258,000	100% (\$3,258,000)
SHORT HILLS-LOOSE CREEK 138kV	Reconductor line	\$598,000	100% (\$598,000)
FOREST GLEN-NEPTUNE 138kV line	Reconductor line	\$1,438,000	5% (\$73,000)
KAW-SPALL NS 138kV	Reconductor line	\$2,753,000	7% (\$202,000)
	Total allocated to project		\$18,066,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.